

① Shankar 18.2.2

$$\vec{E}(t) = \mathcal{E} e^{-t^2/\tau^2} \hat{e}_z \quad \text{so } V = -e \mathcal{E} \underbrace{\hat{r} \cos \hat{\theta}}_{=z} e^{-t^2/\tau^2}$$

$$C_{n=1,0,0}^{n=2,l,m} = \frac{-i}{\hbar} \int_{-\infty}^{\infty} dt e^{i\omega t} e^{-t^2/\tau^2} (-e\mathcal{E}) \langle 2lm | \hat{r} \cos \hat{\theta} | 100 \rangle$$

Recall that $\langle \hat{r} | 100 \rangle$ has no θ, ϕ dependence, and $r \cos \theta = \sqrt{\frac{3}{4\pi}} Y_{10}(\theta, \phi) r$ so $\langle 2lm | \hat{r} \cos \hat{\theta} | 100 \rangle \neq 0$ only for $l=1, m=0$ since everything else is orthogonal to $\hat{r} \cos \hat{\theta} | 100 \rangle$.

$$\langle 210 | \hat{r} \cos \hat{\theta} | 100 \rangle = \int_0^\infty dr r^2 \int_{-1}^1 d \cos \theta \int_0^{2\pi} d\phi R_{21}(r) Y_{10}^*(\theta) r \cos \theta R_{10}(r) Y_{00}$$

$$= \frac{\sqrt{3}}{4\pi} \cdot 2\pi \cdot \int_{-1}^1 dx x^2 \int_0^\infty dr r^3 R_{21}(r) R_{10}(r) \quad \text{where } x \leftrightarrow \cos \theta$$

$$= \frac{1}{\sqrt{3}} \int_0^\infty dr r^3 R_{21}(r) R_{10}(r) = \frac{1}{\sqrt{3}} \int_0^{2a_0} dr r^2 \frac{1}{\sqrt{24}} \frac{1}{a_0^{5/2}} r e^{-r/2a_0} r \frac{2}{a_0^{3/2}} e^{-r/a_0}$$

$$= \frac{1}{\sqrt{12}} \int_0^{2a_0} dr r^4 e^{-\frac{3r}{2a_0}} \frac{1}{a_0^4} = \frac{1}{\sqrt{12}} \frac{4!}{a_0^4} \left(\frac{2a_0}{3} \right)^5 = \sqrt{2} \frac{2^7}{3^5} a_0$$

$$\text{So } C_{100}^{210} = \frac{-i}{\hbar} \sqrt{2} \frac{2^7}{3^5} a_0 (-e\mathcal{E}) \underbrace{\int_{-\infty}^{\infty} dt e^{i\omega t} e^{-t^2/\tau^2}}_{= \sqrt{\pi} \tau e^{-\frac{(\omega\tau)^2}{2}}}$$

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2a) Start w/

$$\textcircled{\bullet} d\psi(t) = \delta\psi i - \frac{i}{\hbar} \int_{t_0}^{t_f} \langle \psi^0 | H'(t') | \psi^0 \rangle e^{i\omega_0 t'} dt'$$

where $H'(t') = V'(t') = \alpha \delta(x - ct')$

$$\begin{aligned} \text{so } \langle \psi^0 | V'(t') | \psi^0 \rangle &= \int dx dx' \langle \psi^0 | X X | V'(t') | X' X' | \psi^0 \rangle \\ &= \alpha \int dx dx' \langle \psi^0 | X \rangle \delta(x - ct') \delta(x - x') \langle x' | \psi^0 \rangle \\ &= \alpha \int_{-\infty}^{\infty} dx \psi_{\psi^0}^*(x) \delta(x - ct') \psi_{\psi^0}(x) \\ &= \alpha \psi_{\psi^0}^*(ct') \psi_{\psi^0}(ct') \end{aligned}$$

so we have

$$\textcircled{\bullet} d\psi(t) = \delta\psi i - \frac{i}{\hbar} \int_{-\infty}^{\infty} \alpha \psi_{\psi^0}^*(ct') \psi_{\psi^0}(ct') e^{i\omega_0 t'} dt'$$

And probability is

$$\boxed{|\psi(t)|^2}$$

$$(b) \quad V(x,t) = \frac{a}{2\pi c} \int_{-\infty}^{\infty} d\omega e^{i\omega(\frac{x}{c}-t)}$$

$$d\psi(t) \text{ is now } \frac{-ia}{\hbar 2\pi c} \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} dx \quad \psi_f^*(x) e^{i\omega(\frac{x}{c}-t)} e^{i\omega_f t} \psi_i(x)$$

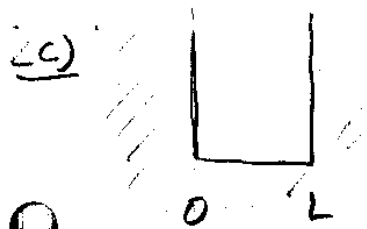
Rearrange integration:

$$\frac{-ia}{2\pi \hbar c} \int d\omega \int dx e^{i\omega \frac{x}{c}} \psi_f^*(x) \psi_i(x) \underbrace{\int dt e^{i(\omega_f t - \omega t)}}_{= 2\pi \delta(\omega_f - \omega)}$$

$$= \frac{-ia}{\hbar c} \int dx e^{i\omega_f x/c} \psi_f^*(x) \psi_i(x)$$

Change variables $x \rightarrow ct'$ $dx = c dt'$

$$= \frac{-ia}{\hbar} \int dt' e^{i\omega_f t'} \psi_f^*(ct') \psi_i(ct') \quad \text{which is same as part a).}$$



$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{otherwise} \end{cases}$$

Solutions are
$$\psi(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right) & 0 < x < L \\ 0 & \text{otherwise} \end{cases}$$

Energies $\Rightarrow E = \frac{n^2 \pi^2 \hbar^2}{2mL^2} ; n = 1, 2, \dots$

Ground state $\psi_0(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$

1st excited $\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right)$

$$\omega_{10} = \frac{E_1 - E_0}{\hbar} = \frac{3\pi^2 \hbar}{2mL^2}$$

So
$$\begin{aligned} \langle f^0 | V'(t') | i^0 \rangle &= \int_0^L \psi_1^*(x) \alpha \delta(x - ct') \psi_0(x) dx \\ &= \alpha \int_0^L \frac{2}{L} \sin\left(\frac{2\pi x}{L}\right) \delta(x - ct') \sin\left(\frac{\pi x}{L}\right) dx \\ &= \alpha \frac{2}{L} \sin\left(\frac{2\pi}{L} ct'\right) \sin\left(\frac{\pi ct'}{L}\right) ; 0 < t' < L/2 \end{aligned}$$

And
$$d_{10}(t) = -\frac{i}{\hbar} \int_0^{L/2} \alpha \frac{2}{L} \sin\left(\frac{2\pi}{L} ct'\right) \sin\left(\frac{\pi ct'}{L}\right) \exp\left(\frac{i 3\pi^2 \hbar}{2mL^2} t'\right) dt'$$

Let $u = \frac{\pi ct'}{L} \Rightarrow -\frac{i 2\alpha}{\hbar} \frac{L}{\pi c} \int_0^{\pi} \sin(2u) \sin(u) \exp(i\beta u) du ; \beta = \frac{3}{2} \frac{\pi \hbar}{mcL}$

$$-\frac{i 2\alpha}{\pi c \hbar} \left[-\frac{4i(1+e^{i\pi\beta})\beta}{9-10\beta^2+\beta^4} \right] = -\frac{8}{\pi c \hbar} \left[\frac{i(1+e^{i\pi\beta})\beta}{9-10\beta^2+\beta^4} \right]$$

$$\text{Prob} = \left| -\frac{8}{\pi c \hbar} \left[\frac{i(1+e^{i\pi\beta})}{9-10\beta^2+\beta^4} \right] \right|^2$$